Molecular hydrogen in EXHALE

Abstract

EXHALE carries H_2 as a single species. The state vector holds one number density for it, the network reacts it as one reagent, and no rovibrational level is transported. Every quantity that depends on how the molecules are distributed over (v, J) therefore has that distribution supplied by assumption, and the assumption is not the same in every place. The infrared line cooling is explicitly LTE and says so in its own function name; the thermal dissociation and its three-body reverse rest on vibrational equilibrium and are now built from one another through the equilibrium constant, so that axis is consistent by construction; the Lyman–Werner photodissociation rate no longer imports a fit at all: it reads a table of a line-by-line calculation, with the lines absorbing each other's beam, run at the temperature and density of this layer, and states where that calculation itself stops being a value; and the H_3^+ cooling, alone among the four, carries an explicit departure factor. This memo records each assumption, the published evidence behind it, the range over which it holds, and where the layer EXHALE actually computes leaves that range.

1. What is and is not carried

H_2 enters the state vector as species `isp_H2` of `src/modules/init/species_table.f90`, one of the four molecular columns (`H2`, `H2p`, `H3p`, `HeHp`) that the molecular option adds. It is written to `output/Ion_species.txt` and `output/Ion_species_adv.txt` as the number density of the molecule; the four molecular columns are the last four of the file, but their absolute position depends on which metal elements the run carries, so they must be located from the `# columns` schema header rather than counted.

What the state vector does *not* carry is any resolution of the rovibrational ladder. There is one H_2 continuity equation, not one per level, and no state-to-state collisional network. Nothing in the code transports a vibrational temperature, an ortho/para ratio, or a level population of any kind. The rovibrational ladder appears twice as *tabulated data*: the partition function of the equilibrium constant (`mol_rates.f90`) and the line list of the infrared cooling (`molecular_infrared_data.f90`), but in both places it is evaluated at the local kinetic temperature and never solved for.

The consequence is the subject of this memo. A rate coefficient measured on one distribution of levels does not transfer to another, and EXHALE uses coefficients measured under at least three different distributions in the same cell. Each of them is defensible where it is used, and the point of collecting them is to say exactly where that is.

1.1 Code map

Table 1 lists every file that touches H_2 and what it does with it. Three groups are worth separating in advance, because the level-distribution assumption enters each of them differently: the files that supply a *rate*, those that supply an *emissivity*, and those that supply a *composition*.

1.2 The layer the assumptions are evaluated in, and what the numbers below are read from

Every claim about whether a fit is used inside or outside its range needs a statement of the conditions EXHALE actually reaches. The reference case here is `mol_lyman_werner` of the regression matrix (the hot-Uranus molecular base with

File	What it does with H_2
`init/`	
`species_table.f90`	Declares `isp_H2` and the nucleus weights (2 H nuclei per molecule) the element budget counts with
`set_IC.f90`	Seeds the H_2 column from the chemical-equilibrium fit at the local pressure and T_0, capped by the available neutral H
`lower_atmosphere/`	
`mol_rates.f90`	The Koskinen et al. (2022) Table-1 network, plus the H_2 thermochemistry: partition function, equilibrium constant, three-body recombination
`lower_column.f90`	`q_h2_equilibrium`, the chemical-equilibrium mixing ratio, and the mean molecular weight of the analytic column built from it
`lyman_werner.f90`, `h2_self_shielding_table.f90`	Solomon-process photodissociation; the absolute dissociation cross section and its two branching ratios tabulated from a line-by-line calculation with overlapping lines, and the Draine & Bertoldi (1996) equivalent width that still sets the band share
`molecular_infrared_ cooling.f90`, `. . . _data.f90`	The H_2 quadrupole and magnetic-dipole line cooling, and the H_2O and CO bands beside it
`h3p_cooling.f90`	H_3^+ infrared cooling; H_2 enters as the collider density of the non-LTE departure factor
`nonlinear_system_solver/`	
`System_HeH_mol.f90`, `System_HeH_mol_metals.f90`	The coupled ionization system: unknown $x(4) = 2n_{H_2}/n_H$, the H nuclei bound in H_2
`constrained_chemical_ equilibrium.f90`	The same network in logarithmic densities, with a continuation in the radiation field seeded from the H_2 dissociation equilibrium
`functions/`	
`composition.f90`	`h2_mixing_ratio_base`, `h2_bound_fraction`, `comp_ntot_bc`: the H nuclei bound into H_2 leave the base particle count
`cross_sec.f90`	`sigma_H2`, the total photoionization cross section above 15.4 eV, and `frac_H2_ dissociative_ionization`, the share of it that leaves $H + H^+$ instead of H_2^+
`binary_element_diffusion.f90`	H_2 as a diffusion carrier of the hydrogen component, with mass 2 and its own polarizability
`radiation/` and `post_process/`	
`util_ion_eq.f90`	`fuv_lw_photon_field`: the shared 912–1110 Å beam, the H_2 column, and the band fraction the pumping lines remove
`util_ion_eq.f90`	`he_rec_coupling`: H_2 competes with H I (and, above 24.6 eV, He I) for the He II recombination photons, and wins essentially all of them in the molecular layer
`Cool_coeff.f90`	`ioniz_HeI23S_H2`, the $He(2^3S) + H_2$ Penning and associative ionization rate
`post_process_adv.f90`	Does not advection-correct the four molecular columns, and says so in its header, but its energy residual does read H_2: the caloric internal energy of the cell and of its upwind neighbor are formed at `h2_particle_fraction` of each (`internal_energy_of_mixture`), so an atomic cell below a molecular one no longer loses the incoming rovibrational energy (§6.4)

Table 1: Where H_2 appears. Paths are relative to `src/modules/`.

`Stellar LW flux [erg/cm2/s]: 343.0`) because it is the only default case that carries the Lyman–Werner channel. Measured on its `output/Lyman_Werner.txt` and `output/Ion_species.txt`:

Layer definition	cells	r/R_p	T [K]	n_H [cm^{-3}]
$x_{H_2} > 0.5$	144	1.0002–1.044	1048–1221	2.03×10^{12}–1.07×10^{13}
$x_{H_2} > 0.01$	184	1.0002–1.075	891–1221	8.49×10^{11}–1.07×10^{13}
$x_{H_2} > 10^{-6}$	476	1.0002–3.712	874–2531	1.69×10^{7}–1.07×10^{13}

with $x_{H_2} = 2n_{H_2}/n_H$ and n_H the total hydrogen nucleus density, over the 500 physical cells only (the innermost ghost at $r = 0.9998$ is excluded); the H_2 density itself peaks at 5.15×10^{12} cm^{-3} in the first physical cell and falls to 4.48×10^{9} cm^{-3} at the 1% edge. The three rows matter because they give three different answers. The widest range, 874–2531 K, is the extent of gas that carries *any* H_2 at all, and its upper half is gas where H_2 is a 10^{-5} trace; the layer where H_2 is actually a constituent spans 874–1221 K. Statements below use the narrow definition unless they say otherwise. (Every number of the table was re-measured on the case output of 2026-09-10, which is the state the goldens now hold; the previous edition of the table, taken on an earlier state of the same case, gave a layer both thicker and about eight times denser at its base.)

These are relaxation snapshots, not converged solutions. Thirteen of the sixteen default regression cases are pinned by a `maxsteps` file and stop there rather than at a convergence criterion: the seven molecular ones and `lower_profile` at 12,000 steps, `oxygen_chemistry` at 1000, `hydrostatic_column` at 300 and the three `hp_*` seeds at 100. `mol_lyman_werner` ends at du = 0.785, i.e. a 78.5% radial spread in the mass flux $\rho v r^2$, well short of the 10^{-3} threshold a converged run meets; only the three `wasp` cases run to a stop of their own, and only `wasp_full_newton` certifies. Every profile number in this memo therefore describes a relaxation snapshot. That is adequate for the purpose (the question is which decade of density and temperature a fit is being asked about, not the fourth digit), but it is not adequate for quoting a mass-loss rate, and none is quoted here.

The snapshots are the goldens. The numbers above come from the case output directories, and those outputs now agree with `backup/regression/golden/`: `mol_lyman_werner`'s `Hydro_ioniz.txt` and `Ion_species.txt` were compared line for line against the snapshot and are identical. This paragraph replaces one saying that every default case differed from its golden, which was true while the goldens predated a series of uncommitted source changes; the goldens were re-snapshotted on 2026-09-10, and `lower_profile` again the same day after the element-projection mass fix.

2. The level-distribution assumptions

2.1 Infrared line cooling: LTE, stated explicitly

Implementation note (2026-10-27). The critical-density comparisons in this memo use the older Hollenbach–McKee coefficients and are stale for the current implementation. The H, H_2, and He collider coefficients are now defined in `h2_vibrational_relaxation.f90`. The relevant LTE condition is that the local sum of collisional de-excitation rates

greatly exceed radiative decay. The quoted critical densities, including the later decision table, must be recomputed with those current coefficients and the local collider mixture before they are used to claim LTE. This comment-review task did not rerun that calculation; the historical values below are retained as records.

`molecular_infrared_cooling.f90` evaluates the H_2 line emission as a Boltzmann sum,

$$E_{H_2}(T) = \sum_l f_u(T) A_l \, \Delta E_l, \qquad f_u(T) = \frac{g_u \, e^{-T_u/T}}{Z(T)}, \qquad Z(T) = \sum_{\mathrm{lev}} g \, e^{-T_{\mathrm{lev}}/T}, \tag{1}$$

and names the routine that returns it `h2_line_emission_lte`. There is no ambiguity about the assumption here; the question is only where it holds.

The line list is Roueff et al. (2019), A&A 630, A58, table 2 (electric quadrupole plus magnetic dipole), and the level ladder comes from the same table. As carried in `molecular_infrared_data.f90` it is `n_h2_lev` = 302 levels and `n_h2_line` = 1833 lines.

Rotational levels are not the constraint: they thermalize at densities far below anything in this layer. The vibrational levels are. The code records the comparison at its own site. The critical density of $v = 1$ is $n_{\mathrm{crit}} = A_{10}/\gamma_{10}$ (Hollenbach & McKee 1979, ApJS 41, 555, p. 576), and with their $A_{10} = 8.3 \times 10^{-7}$ s^{-1} (p. 583, J-averaged, from Turner, Kirby-Docken & Dalgarno 1977) and the rate coefficients of their eq. (6.29) it is 2×10^6–6×10^6 cm^{-3} over 900–1350 K with H_2 as the collider and 5×10^4–8×10^4 cm^{-3} with atomic H. The measured base density of the molecular gate case, $n(H_2) = 5.15 \times 10^{12}$ cm^{-3}, is six to eight decades above either.

Two caveats on that comparison. Neither Hollenbach & McKee nor Burton, Hollenbach & Tielens (1990) gives a *helium* collision rate for H_2 $v = 1$ (their colliders are H, H_2 and, in the latter, electrons), so a helium-dominated gas is outside both sources; omitting helium from the collider sum understates the de-excitation rate and therefore errs against claiming LTE, not for it. And the figure quoted here previously, 10^9–10^{11} cm^{-3}, carried no citation and is three to five decades too high; correcting it widens the margin rather than narrowing it.

Above the base the density falls, but so does the temperature, and the vibrational bands then carry a small share of the emission. That is the quantitative statement that makes the LTE overestimate harmless where LTE is least justified, and it can be evaluated directly from the line list: sum the emission of the lines whose upper level has $v \geq 1$ and divide by the total. Table 2 does so.

T [K]	$E_{H_2}(T)$ [erg s^{-1} molecule^{-1}]	share from $v_u \geq 1$
400	3.034×10^{-23}	0.72%
500	7.717×10^{-23}	5.70%
600	1.789×10^{-22}	18.2%
1000	2.726×10^{-21}	65.7%

Table 2: Share of the LTE H_2 line emission carried by transitions out of a vibrationally excited upper level, evaluated on the implemented line list. Outside the tabulated range the share is 0.016% at 300 K and 81.5% at 2000 K. The level table carries no (v, J) index, so the vibrational assignment was recovered from the Roueff et al. (2019) source table and matched back onto the 1833 selected lines; classifying instead by $v_u \neq v_l$ (a band transition rather than a pure rotational one) changes the shares by less than 0.1 percentage point up to 1000 K.

The reading is that the LTE assumption is safe at the base, where the density supports it, and that where the density no longer supports it the vibrational bands (the levels the density fails to thermalize) have already stopped carrying the emission. Below about 500 K the error LTE can make is bounded by a few percent of the total, because that is all the vibrational levels contribute.

2.2 Thermal dissociation and three-body recombination: vibrational equilibrium

The pair R12 (H_2 + M → H + H + M) and R15 (H + H + M → H_2 + M) is one channel run in two directions. Since the change recorded below, `mol_rates.f90` builds R12 from R15 through the equilibrium constant,

$$k_{\rm diss}(T) = \frac{k_{\rm rec}(T)}{K_{\rm eq}(T)}, \qquad K_{\rm eq}(T) = \frac{n_{\rm H_2}}{n_{\rm H}^2}\ [\rm cm^3], \tag{2}$$

rather than transcribing an independent dissociation fit.

That construction is what makes this axis consistent. $K_{\rm eq}$ is built in `h2_thermochemistry_init` from the translational partition functions and the *rovibrational* internal partition function `q_rovib_H2`, summed over the Morse ladder of H_2 $X\,^1\Sigma_g^+$ with the ortho/para nuclear-spin weights. It therefore contains vibrational excitation explicitly: both sides of the equilibrium refer to a vibrationally equilibrated gas, and the reverse rate built from it inherits that and nothing else.

The measured coefficient it is anchored on refers to the same state, and the source says so. Breshears & Bird (1973), 14th Symposium (International) on Combustion, 211, read their initial rate off the minimum of the postshock density-gradient dip, which they identify as "the rate corresponding to conditions of vibrational equilibrium in the thermal sense but with negligible extent of dissociation". The same paper bounds where its rate law applies: the "observation of induction periods for diatomic dissociation clearly indicates that the assumptions implicit in Eqs. (2)–(4) are not valid during the very early course of the reaction; nevertheless, they may be applicable during the 'steady state' which follows the induction period." A hydrodynamic model that resolves nothing faster than a vibrational relaxation time is in that steady state throughout. Their Eq. (3), $k_d/k_r = K_{\rm eq}$, states the detailed-balance relation outright, so deriving one direction from the other is the relation the measured coefficient already carries rather than an assumption added here.

The critical density for vibrational thermalization, 10^4–10^8 cm^{-3} depending on the collider, sits many orders below the 10^{13}–10^{15} cm^{-3} of the base, which is the independent check that the premise holds where the rate is evaluated.

Status of this change. R12 as detailed balance is present in the working tree and *not* in the last commit: `git show HEAD:src/modules/lower_atmosphere/mol_rates.f90` still carries the independent Baulch et al. (1992) fit $1.5 \times 10^{-9} e^{-48350/T}$, and the changelog carried no entry for the change. It is therefore behind the code on this point.

2.3 Lyman–Werner self-shielding: no longer an imported fit

This was the weakest of the four assumptions, and it is the one that has been replaced by a calculation rather than better-argued. `lyman_werner.f90` took the self-shielding factor from Draine & Bertoldi (1996), ApJ 468, 269 (hereafter DB96), their eq. (37); for a time it took it from the temperature-dependent fit of Richings, Schaye & Oppenheimer (2014) instead. Both are closed forms of the same two-term shape, both were built for interstellar gas, and neither had ever been tested at the temperature of a planetary molecular base.

What was done instead. Both fits were measured against a level-resolved CLOUDY c25.00 calculation run on our own grid: T = 700–3200 K at $n_{\rm H} = 10^{12}$–10^{14} cm^{-3}, columns to 5×10^{21} cm^{-2}, pure H/He, no dust, thermal Doppler width, and our own band flux, and neither survived it. Over $10^{18} \leq N_{\rm H_2} \leq 5 \times 10^{20}$ cm^{-2}, the columns where our H_2 sits, DB96 runs from 1.85× high at 900 K to 2.3× low at 2700 K, and Richings is low by 3.3–4.3× at every temperature of the layer. The failure is one of *shape* and not of level: the calculated $f_{\rm shield}(N_{\rm H_2})$ has a trough near 10^{16}–10^{17} and a shoulder near 10^{19} cm^{-2} that a two-term algebraic form of this kind has no freedom to place, so no rescaling of either fit could have worked.

And then the CLOUDY calculation was replaced as well (2026-09-03). It carried two errors of opposite sign that crossed near $N_{\rm H_2} = 2 \times 10^{20}$ cm^{-2} and cancelled below it. First, the quantity tabulated was CLOUDY's `Shield(H2)` column, which is the dissociation rate of H2g alone (the sub-reservoir $v = 0$, $J \leq 8$ that CLOUDY splits off at `ENERGY_H2_STAR` = 4100 cm^{-1}) divided by the H2g density, while EXHALE carries one H_2 species and applied it to all of it. The vibrationally excited part starts from levels 10^2 to 10^3 times less populated, saturates far later, and keeps dissociating after the H2g lines have gone black; weighting the two reservoirs by their own densities raises the rate by up to a factor 11 at $N_{\rm H_2} = 10^{19}$ cm^{-2}. Second, CLOUDY applies the Federman, Glassgold & Kwan (1979) shielding function to each transition using that transition's own optical depth, so no line absorbs any other line's beam. Sec. 135 of `Update_EXHALE` and `md/p38_line_overlap_shielding.md` are the measurement.

What is tabulated now is a line-by-line calculation of the same three quantities on the same $39 \times 9 \times 3$ grid: a plane-parallel slab, a normally incident beam, flat F_λ normalized to F = 343 erg cm^{-2} s^{-1} over 912–1110 Å, and $\tau(\nu, N) = N \sum_i x_i \sigma_i(\nu)$ summed over every Lyman and Werner transition between 911.75 and 1200 Å on a 4×10^5-point frequency grid, so every line is shielded by every other line. The profiles are Voigt with $b = (2kT/m_{\rm H_2})^{1/2}$, the populations LTE at the imposed T, and the line data are Abgrall, Roueff & Drira (2000), the same measurements CLOUDY's `dissprob_*` and `transprob_*` files and the Meudon PDR code carry, so no line-list difference enters the comparison. `h2_self_shielding_table.f90` holds $\log_{10}\sigma_{\rm diss}$, $\log_{10} p_{\rm single}$ and $\log_{10} p_{\rm eff}$, trilinear in $(\log_{10} T, \log_{10} n_{\rm H}, \log_{10} N_{\rm H_2})$, with every axis clamped rather than extrapolated. The generator is `src/utils/h2_shielding_table_line_by_line.py`; `md/h2_self_shielding_cloudy.md` and `src/utils/h2_shielding_table_from_cloudy.py` record the CLOUDY runs, which still supply the fluorescent trapping.

The one piece that is not line-by-line. The trapping ratio $p_{\rm eff}/p_{\rm single}$ (the suppression of the branching by re-absorption of the fluorescent decay photons) is read node by node from the same 27 CLOUDY runs and multiplied in, rather than reimplemented as a second and unvalidated version of the same thing. Its geometry is plane-parallel, illuminated on one face and closed on the other. Our layer has the same thick-inward side, which is the side that matters, but it is spherical and open outward. This is the only ingredient of the table that carries a geometry our layer does not have, and it is an open item.

What the level distribution now is. It is an LTE population at the imposed T, not one imported from a 10^2 K interstellar front, and the assumption is checked rather than asserted: at $n_{\rm H} \geq 10^{12}$ cm^{-3} the collisional rates exceed the pumping rate by orders of magnitude, and the CLOUDY runs on the same grid, which solve the populations, return

departure coefficients of 1.000 at every level. That is what our base is, and it is precisely the population no published fit was built on. The open item this subsection used to carry, "a calculation of $f_{\rm shield}$ on a 10^3 K collisionally equilibrated population would settle whether the fit transfers", has been done, and the answer was that it does not.

Three axes, now measured rather than assumed.

- **Density.** The line-by-line pumping has no density in it at all: the populations are LTE, so the tabulated $\sigma_{\rm pump} = \sigma_{\rm diss}/p_{\rm eff}$ is identical to every stored digit across $n_{\rm H} = 10^{12}$–10^{14} cm^{-3}. The axis survives only because the trapping ratio came from CLOUDY, and through it $f_{\rm shield} = \sigma_{\rm diss}(N)/\sigma_{\rm diss}(N_{\rm min})$ moves by at most 2.5% below $N_{\rm H_2} = 10^{19}$ cm^{-2} and by up to 23% at 10^{21}: measured on the installed table. The absence of a density argument in either published fit was never what was wrong with them.
- **Radiation field.** One incident flux only, $F = 343$ erg cm^{-2} s^{-1} over the band. With LTE populations the line-by-line shielding cannot depend on it; in the CLOUDY runs, where the populations were solved, dropping the band flux by four decades (G_0 from 1.25×10^6 to 125) moved $f_{\rm shield}$ by at most 9% anywhere and by under 5% below $N_{\rm H_2} = 10^{20}$. There is no radiation-field axis to carry.
- **Doppler parameter.** Thermal only, $b = (2kT/m_{\rm H_2})^{1/2}$, 2.7–4.7 km s^{-1} over the grid: the same expression the code's own `h2_doppler_parameter` evaluates, so the table and the code agree on it by construction. The table needs no b argument: b is a function of T, which is already an axis.

Where the table stops being a value, and why that is the honest statement. There is no longer a column above which the tabulated factor is an upper bound. Line overlap (what DB96 §5.2 say produces "the rapid falloff … for $N_2 > 10^{20}$ cm^{-2}") is inside the calculation, and at the columns a planetary molecular base reaches it is not a correction but the dominant term. Two calculations differing only in whether the lines absorb each other's beam give, at 1300 K, ratios of 0.62 at $N_{\rm H_2} = 10^{19}$, 0.33 at 10^{20}, 0.042 at 10^{21} and 0.0016 at 4.4×10^{21} cm^{-2}.
`h2_shielding_overlap_column`, which returned the column at which the summed equivalent width of the pumping lines reached the whole band (1.4–1.8 $\times 10^{21}$ cm^{-2} at 700 K falling to 2.5×10^{20} at 3200 K), is gone from the module together with the plane it interpolated. The boundary it drew was about a decade optimistic: at the column it returned, the one-line-at-a-time table already stood 7 to 10 times above the overlapping-line answer.
What replaces it is a grid boundary and nothing more. `h2_shield_max_column()` returns the top of the tabulated column axis, 5.0×10^{21} cm^{-2}, above which the edge value is used, and the two run-time warnings that quoted the overlap column now report that extrapolation instead. Our base sits at $N_{\rm H_2} = 4.4 \times 10^{21}$, inside the axis; only the transient excursion to 5.3×10^{21} crosses it.

What the calculation was checked against. At 100 K it reproduces the DB96 self-shielding function to 2–9 percent over $10^{16} \leq N_{\rm H_2} \leq 10^{21}$ cm^{-2}. That fit rests on a calculation which includes overlap (their §5.2), and it is the one published overlap-including curve there is to check against; the same calculation run one line at a time departs from DB96 exactly where DB96 say overlap sets in, by 3.3× at 10^{21} and 12× at 4.4×10^{21}. Against a second code that carries overlap: the Meudon PDR code (Le Petit et al. 2006) with its exact UV line transfer, 30 H_2 levels and 20 global

iterations on pure H/He at $n_{\rm H} = 10^{13}$ cm^{-3}: the two agree to 5–22 percent over $10^{19} \leq N_{\rm H_2} \leq 10^{21}$ cm^{-2} at 900, 1300 and 1800 K. At the illuminated face, the lowest tabulated column $N_{\rm H_2} = 10^{12}$ cm^{-2}, the absolute quantities are $\sigma_{\rm pump} = 2.59 \times 10^{-17}$ (700 K) to 2.88×10^{-17} cm^2 (3200 K) against DB96's own $3.4520 \times 10^{-18}/0.135 = 2.557 \times 10^{-17}$ cm^2, and $p_{\rm single} = 0.143$ to 0.159, which at 1300 K is 0.1475 against 0.1465 from the CLOUDY runs. The Fortran table reproduces the Python calculation at all 1053 grid points to 3.1×10^{-9} relative on $\sigma_{\rm diss}$.

The rate is the tabulated cross section itself (2026-09-02). The module no longer multiplies a constant cross section by this factor. What is tabulated is $\sigma_{\rm diss}(N_{\rm H_2}, T, n_{\rm H})$, the dissociation cross section per *incident* band photon, with the self-shielding and the trapping of the fluorescent decay photons both inside it; $f_{\rm shield}$ survives as the ratio $\sigma_{\rm diss}(N)/\sigma_{\rm diss}(N_{\rm min})$, a diagnostic. The earlier reading of the same runs (that $\sigma_{\rm LW}$ was 23–60% low and $\langle p_{\rm diss} \rangle$ 31% high) was a comparison of two different quantities: the line data agree to 0.8 percent, and the gap is line trapping. One absorption has *two* branching ratios and the code now carries both from the same table: $p_{\rm single}$, the untrapped $D/(D+\sum A_{ul})$, which fixes how many fluorescent decays accompany each dissociation, and $p_{\rm eff}$, which fixes how many band photons the beam loses per dissociation. Sec. 132 of `Update_EXHALE` and `md/p39_lw_cross_section_sources.md` carry the derivation; Sec. 135 and `md/p38_line_overlap_shielding.md` carry the rebuild of the table these three are read from.

What DB96 still owns. The band share A that H_2 removes from the shared 912–1110 Å beam is DB96 eq. (39), the closed-form integral of their own eq. (37), and it stays there. Its normalization is the one quantity in this module that no line transfer touches (the *pump* cross section), and there the two sources agree: 2.640×10^{-17} cm^2 at 1300 K at the illuminated face of the line-by-line calculation against 2.557×10^{-17} from DB96's own Table 1–2, 3 percent.

The energy the fluorescence leaves behind is computed, not adopted. Burton, Hollenbach & Tielens (1990) adopt 2.0 eV per pump in one sentence of their Appendix A, without a derivation. The same quantity follows from the Abgrall, Roueff & Drira (2000) transition probabilities: summing every decay of every in-band pump, weighted by the $X(v'', J'')$ energy it lands on, gives a mean landing energy of 2.060 eV at 700 K rising to 2.129 eV at 3200 K per *bound* fluorescence, 3 to 6 percent above the adopted value. `h2_energy_per_bound_fluorescence_eV(T)` replaces the constant, and the photoelectron channel of `electron_energy_degradation.f90` shares it with its own state-resolved branching. The average is over unshielded line weights; with shielded weights it is 2.17–2.25 eV, so inside the layer the term is a lower bound by about as much as the correction it applies.

2.4 H_3^+ cooling: a departure factor, not LTE

The one place where EXHALE does not assume LTE is the H_3^+ infrared cooling, and it is worth stating because it runs the other way from §2.1. `h3p_cooling.f90` multiplies the LTE emission $E(T)$ of one H_3^+ molecule, the MiZATeP cooling function of Mizus et al. (2017, tabulated 20–5000 K; it replaced the Miller et al. (2013) LTE fit on 2026-10-10, update log stage 3 section 132, because that fit lies above it by 1.10–2.1 at 900–3000 K), by the departure factor $s(T, n_{\rm H_2})$ of Miller et al. (2013) Table 6, interpolated as a power law in $n_{\rm H_2}$ between its columns and linearly in T between its rows; below 10^6 cm^{-3} the collisional limit $s \propto n_{\rm H_2}$ is used and above 10^{14} cm^{-3} the table edge ($s = 0.9985$–

1) is held (`h3p_nonlte_factor`). Here H_2 enters as the collider density that sets how far the H_3^+ levels depart from a Boltzmann distribution.

The net exchange is written in Kirchhoff form,

$$\Lambda_{\rm net} = n_{H_3^+}\, 4\pi\, s(T, n_{H_2})\, [E(T) - W\, E(T_{\rm rad})]\,, \tag{3}$$

with W the dilution of the field the unresolved lower atmosphere presents, switched on by the input key `Base IR field` (default `False`). The departure factor multiplies both terms, which it must: $s < 1$ applied to the emission alone would heat a gas hotter than the field, and the form above is exactly zero at $T = T_{\rm rad}$, $W = 1$.

The factor is doing real work rather than sitting at its LTE value: evaluated over the $x_{H_2} > 0.01$ layer of `mol_lyman_werner`, s runs from 0.996 in the base cell down to 0.438 at the top of that layer. Assuming LTE there would overstate the H_3^+ cooling by a factor 2.3. *Those two values of s are STALE*: they were taken on an earlier state of the case, whose base carried $n_{H_2} = 4.63 \times 10^{13}$ cm^{-3} and whose 1% edge sat at 4.42×10^9 cm^{-3}, $r = 1.172\, R_{\rm p}$; the state the goldens now hold spans 5.15×10^{12} down to 4.48×10^9 cm^{-3} over 1.0002–1.075 $R_{\rm p}$ (§1.2). Re-measuring them means evaluating `h3p_nonlte_factor` of `h3p_cooling.f90` on the two ends of that layer; the density range still sits inside the Miller et al. Table 6, so the verdict of Table 3 does not turn on it.

The table is a vacuum departure factor (collisions against spontaneous decay, no incident field), so a field strong enough to matter pumps the levels back toward LTE and the true departure is smaller than s. The treatment therefore underestimates the magnitude of the exchange, not its sign or its fixed point.

For comparison, Frelikh & Murray-Clay (2026), ApJ 996, 96 treat the same coolant in pure LTE. On this one channel EXHALE carries the more complete physics.

2.5 The chemical-equilibrium mixing ratio

`q_h2_equilibrium` (`lower_column.f90`) is the Koskinen et al. (2022) Eq. (11) fit to the Visscher et al. (2006) chemical-equilibrium H_2/H partition,

$$q_{H_2} = \frac{1.9845 + 10^u - \sqrt{10^u (3.9690 + 10^u)}}{2.3670}, \qquad u = -\frac{23672}{T} - \log_{10} p[\text{bar}] + 6.2645, \tag{4}$$

with limits $1.9845/2.3670 = 0.83840$ as $u \to -\infty$ (fully molecular, the fit's approximation of the solar-composition volume mixing ratio $0.5/(0.5 + \text{He/H}) = 0.863$) and 0 as $u \to +\infty$.

It is used in four places: the mean molecular weight of the analytic lower column, the base H_2 mixing ratio when no `base.inp` supplies a photochemical one, the initial-condition seed of the H_2 column, and the $\lambda \to 0$ end of the continuation path in `constrained_chemical_equilibrium.f90`.

The assumption here is not about level populations but about the chemistry: this is *thermochemical* equilibrium, and it knows nothing about photodissociation. It is therefore not a statement about an irradiated wind, and the code says so at its own site: in chemical equilibrium $q_{H_2}(1\,\mu\text{bar})$ stays large up to $T \sim 2000$ K and the gas is fully atomic only above ~ 2400 K, while photochemistry plus the hotter actual base temperature dissociate H_2 well below that. Using it as a *seed* for a solve that then applies the full radiation field is exactly the use it supports; using it as a discriminant for whether a cell should be molecular is not. The changelog records an attempt in that direction being withdrawn, together

with the repair of an asymptotic guard in the same function that had been inverted since transcription and handed every cell below about 529 K a fully atomic state labeled “the dense molecular limit”.

2.6 Three sites that inherit a ground-state assumption silently

Three further quantities are measured on ground-state H_2 and used without a level-distribution correction. None is flagged in the code as such, and none is likely to matter at the level the three above do, but the list should be complete.

- **Photoionization.** `sigma_H2` in `functions/cross_sec.f90` is the Yan, Sadeghpour & Dalgarno (1998), ApJ 496, 1044 fit (their Eqs. 17–19), threshold 15.4 eV, with the near-threshold resonance structure smoothed as in the source fit. It is a cross section for H_2 in its ground vibrational state; a vibrationally hot molecule has a lower threshold and different near-edge structure. The branching between its two product channels (`frac_H2_dissociative_ionization`, Sec. 3.) carries the same assumption: it is measured on a room-temperature target.
- **Diffusion.** `binary_element_diffusion.f90` gives H_2 the static dipole polarizability $\alpha = 5.315\,a_0^3$ (Schwerdtfeger & Nagle 2019) for the ion-neutral coefficient. This is the $v = 0$ value; polarizability rises with vibrational excitation.
- **Metastable-helium Penning ionization.** `ioniz_HeI23S_H2` in `radiation/Cool_coeff.f90` is the Maxwell–Boltzmann average of the Cohen & Lane (1977) cross sections, taken from the Garcia Munoz (2025) network file, valid over 500–10^4 K following its tabulation. It is a thermal average over relative translational velocity with the internal state of the H_2 fixed.

3. The two product channels of the H_2 photoionization

A photon absorbed by H_2 above threshold leaves one of two things behind, and the network treats them as different reactions because they are:

$$\text{(a)}\quad H_2 + h\nu \;\rightarrow\; H_2^+ + e^-, \qquad E_{\rm th} = 15.4\ \text{eV}, \tag{5}$$

$$\text{(b)}\quad H_2 + h\nu \;\rightarrow\; H + H^+ + e^-, \qquad E_{\rm th} = 18.076\ \text{eV}. \tag{6}$$

Channel (b), *dissociative* photoionization, destroys the molecule without making H_2^+, so it does not feed R8 ($H_2^+ + H_2 \rightarrow H_3^+ + H$) and does not reach the H_3^+ cooling at all; what it produces is a proton and an H atom. In the shielded molecular base, where the H I photoionization that normally supplies protons is starved, it is the dominant photon-driven proton source: measured on the converged He/H = 0.0793 hot Uranus, $P_{\mathrm{H_2,di}}\, n_{\mathrm{H_2}}$ is 21 times $P_{\rm HI}\, n_{\rm HI}$ at $1.004\,R_p$ and falls through unity near $1.04\,R_p$.

`sigma_H2` is the *total* photoionization cross section, so (b) is a branching of it and not an addition: the H_2 destruction rate is unchanged, and only the products move. The branching is

$$f_{\rm di}(E) \;=\; \frac{\sigma(H^+)}{\sigma(H^+) + \sigma(H_2^+)},$$

tabulated at the 69 energies of Chung, Lee, Masuoka & Samson (1993), J. Chem. Phys. 99, 885, Table II, and interpolated linearly in E. It rises within 0.03 eV of threshold to a plateau near 0.02, carries the Q_1 Rydberg autoionization resonance (0.108 at 35 eV, back to 0.091 at 37.5 eV), peaks at 0.222 at 74 eV, and is held at its 124 eV value 0.2048 above the top of the measured range. The rows of the coupled system this drives are (4) H_2, which sees the whole rate; (5) H_2^+, which sees $P_{H_2} - P_{H_2,di}$; and (1) H^+, which gains $P_{H_2,di}\, n_{H_2}$. The H atom of the pair is supplied by the H-nucleus closure, exactly as the two Lyman–Werner fragments are.

The secondary electrons do the same thing: Dalgarno, Yan & Liu (1999) give the dissociative channel a mean energy per ion of $22\,W_m(H_2^+)$, so the fast electrons make one such proton for every 22 H_2^+ ions, an additional yield rather than a share of the H_2^+ one.

Yan et al. (1998) misdescribe this branching, and the code does not follow them. Their Sec. 4 gives the dissociative "ratio to the total photoionization cross section" as 0.284 at 76 eV and 0.258 at 124 eV. Those are Chung et al.'s H^+/H_2^+ *column*, a ratio to the H_2^+ partial cross section; the ratio to the total is $r/(1+r)$, i.e. 0.221 and 0.205. Using the quoted numbers as a branching of `sigma_H2` overstates the channel by a factor 1.26–1.28 above 70 eV and understates it by up to 20× below 20 eV.

4. The third body

R12, R13 and R15 are three-body reactions written as a two-body coefficient times a density. Koskinen et al. (2022) print that density as "n" and do not say which particles it counts; their models are H_2/H-dominated Jovian and Neptunian envelopes, where M is H_2 and H. EXHALE passes the total gas-particle density with electrons excluded (`ion_cell_state%ntot`), so every heavy particle is treated as an equally efficient third body.

The coefficient this assumes uniform is the M = H_2 one: R15 carries Cohen & Westberg's (1983) recommended $k_1(H_2) = 2.8 \times 10^{-31} T^{-0.6}$ cm^6 s^{-1} over 50–5000 K, labeled M = H_2 on their data sheet and separate from their k_1(H).

Breshears & Bird measured all three of their colliders in one experiment with one analysis, so their dissociation coefficients are the clean statement of how much the third body matters:

$$\begin{aligned} k_d(\mathrm{M=Ar}) &= 1.55 \times 10^{-10}\, e^{-44736/T}, \\ k_d(\mathrm{M=H_2}) &= 5.48 \times 10^{-9}\, e^{-52989/T}, \qquad [\mathrm{cm^3\,s^{-1}}] \\ k_d(\mathrm{M=H}) &= 3.52 \times 10^{-9}\, e^{-43881/T}. \end{aligned} \tag{7}$$

Evaluated here, $k_d(H)/k_d(H_2)$ is 8.67 at 3500 K, 6.26 at 4000 K, 3.97 at 5000 K and 2.01 at 8000 K: a factor of a few across their measured 3500–8000 K range. Cohen & Westberg's independent *recombination* pair, $k_1(H) = 8.8 \times 10^{-33}$ against $k_1(H_2) = 2.8 \times 10^{-31} T^{-0.6}$, gives 4.21 and 5.21 at 3500 and 5000 K, which is the agreement detailed balance requires of two independent sources inside their ranges.

Below the measured range the two Arrhenius forms diverge, because their activation temperatures differ by 9108 K: the ratio reaches 13.4 at 3000 K, 61.0 at 2000 K and 5.80×10^3 at 1000 K, against 1.98 from Cohen & Westberg at 1000 K. Take a factor of 2 to 5 as the collider spread where H_2 exists, and read the 5.8×10^3 as the cost of extrapolating an Arrhenius fit rather than as physics.

Helium is the collider nobody measured. The equal-efficiency assumption is weakest in two places, and only one of them is quantified above. Above the H_2 front atomic H dominates the collider population, and the factor of 2–5 is the size of the error there: carried, at least, by a vanishing H_2 density. In a helium-rich envelope M is helium, and there is no measurement to compare against at all. Breshears & Bird shocked H_2 in argon and in xenon (5.0, 10.0, 20.0% H_2 in Ar; 5.0, 10.0, 20.2% in Xe) and extracted M = Ar, Xe, H_2 and H; helium appears in their paper only as a shock-driver gas and in "helium–neon laser". None of the sources behind R12, R13 or R15 supplies a helium efficiency, and none is applied. On general grounds a monatomic third body has no internal modes to take up the released energy and should be *less* efficient than H_2, so in a helium-dominated base these three rates are upper bounds; the size of a plausible factor is of order 0.5, a guess rather than a measurement. Nothing in this repository supports a helium efficiency near unity, and no such number should be quoted from it.

5. The caloric equation of state of the molecular layer

The thermal equation of state of the code is composition exact: the pressure is $p = (n_{\mathrm{tot}} + n_e)kT$ with the species count of the cell. The caloric half, the relation between the internal energy density and the pressure, was not. It used the monatomic $e = p/(\gamma - 1)$ with $\gamma = 5/3$ everywhere, which asserts that every particle stores $\frac{3}{2}kT$.
Molecular hydrogen does not. Its rotational constant gives $\theta_{\mathrm{rot}} = 85.35$ K, so at the 400–1300 K the molecular layer runs the rotational ladder is fully excited and an H_2 molecule stores about $\frac{5}{2}kT$; the vibrational ladder adds to that above ~ 800 K. In a gas that is 80 percent H_2 the specific heat is therefore wrong by a factor 1.67.

5.1 The mixture

For a mixture in which every atom, ion and free electron carries $\frac{3}{2}k$ and H_2 carries its own rovibrational heat capacity, the internal energy density is

$$e \;=\; \tfrac{3}{2}\,(n_{\mathrm{tot}} + n_e)\,kT \;+\; n(\mathrm{H_2})\,k\,u_{\mathrm{rv}}(T), \tag{8}$$

with u_{rv} the mean rovibrational energy of one H_2 molecule in units of k (so u_{rv} has the dimension of a temperature). Because $p = (n_{\mathrm{tot}} + n_e)kT$ is unchanged, the exponent that makes $e = p/(\gamma_{\mathrm{eff}} - 1)$ exact is

$$\gamma_{\mathrm{eff}} \;=\; 1 + \frac{n_{\mathrm{tot}} + n_e}{\frac{3}{2}(n_{\mathrm{tot}} + n_e) + n(\mathrm{H_2})\,c_{\mathrm{rv}}(T)}, \qquad c_{\mathrm{rv}} = \frac{\mathrm{d}u_{\mathrm{rv}}}{\mathrm{d}T}. \tag{9}$$

γ_{eff} is a function of the composition *and* of the temperature, and it is not a free parameter: both quantities in (8) are already carried by the composition module.

5.2 Where the level data comes from

u_{rv} and c_{rv} are evaluated from the complete bound rovibrational ladder of H_2 $X\,{}^1\Sigma_g^+$ (302 levels reaching 51 966 K, the 4.478 eV dissociation limit), which the code already carries for its infrared line cooling (Roueff et al. 2019). The level term energies and the weights $g_I(2J+1)$ come from that same table, so there is no second copy of the H_2 spectroscopic constants in the code to drift out of step with the first.

Using the observed ladder is not a refinement for its own sake. A rigid rotor plus a harmonic oscillator, with $\theta_{\mathrm{rot}} = 85.35$ K and $\theta_{\mathrm{vib}} = 5987$ K, gives $c_v(\mathrm{H_2})/k$ smaller by 1.4 percent at 200–900 K, 2.3 percent at 1300 K and 7.3 percent at 3000 K.

Because the sum runs over a single partition function with the nuclear-spin weights inside it, (8) is the ortho/para *equilibrium* mixture. A frozen 3 : 1 mixture gives the same c_{rv} to four decimals at and above 300 K and differs only below ~ 250 K, which is colder than the layer reaches, so the distinction does not arise here.

$\mathrm{H_2^+}$, $\mathrm{H_3^+}$ and $\mathrm{HeH^+}$ are given $\frac{3}{2}k$ each, as if monatomic. Measured over every cell in which $\mathrm{H_2}$ exceeds 10^{-3} of the particle count, the largest ratios to $\mathrm{H_2}$ are 1.7×10^{-7} ($\mathrm{H_2^+}$), 1.2×10^{-7} ($\mathrm{HeH^+}$) and 3.2×10^{-8} ($\mathrm{H_3^+}$), so giving all three the full $\mathrm{H_2}$ heat capacity would move C_V by less than 10^{-6}.

5.3 Inverting it

With the vibrational ladder in c_v, the map from internal energy to temperature stops being a multiplication. It is inverted exactly, by a bisection-safeguarded Newton iteration on the bracket $0 < T \leq u/1.5$, which holds because u_{rv} is non-negative and increasing in T. It is *not* lagged at a previous temperature: a residual that depended on the previous state would blur both the fixed point of the steady solver and the finite-difference Jacobian it is found with.

To keep the cost of that iteration bounded, u_{rv} is tabulated once on a 4096-point logarithmic grid from 1 K to 5×10^4 K and interpolated by a cubic Hermite in $\ln T$ built on exact nodal values of both u_{rv} and $\mathrm{d}u_{\mathrm{rv}}/\mathrm{d}\ln T$. The interpolant is therefore C^1 and its own derivative (the quantity the Newton iteration differentiates) is exact at the nodes. Against the direct 302-level sum over 200–5000 K the largest relative error is 10^{-12} in u_{rv} and 9×10^{-10} in c_{rv}.

5.4 What carries the local index, and what does not

The physical content of γ sits entirely in the $E \leftrightarrow p$ conversion, and that is where the local value is used: the primitive/conservative conversions, the geometric source, the total energies and the physical flux of the Riemann solver, the low-Mach dissipation pressure, the semi-implicit energy update, the heat capacity of the conduction operator, the state scales of the steady solver, and the advected energy equation of the post-processor. The sound speeds likewise use each cell's own γ_{eff}.

The reconstruction needs no index at all: it works in primitive variables. Each face state's energy and sound speed are built with the heat capacity of the cell it was reconstructed from, which is well defined because the two stored composition quantities are ratios of number densities to ρ (functions of the mass fractions alone), and so are unchanged by the density difference between a cell average and its face value.

One place still needs a single index for both sides: the Roe average $a_{\mathrm{avg}} = \sqrt{(\gamma-1)(H_{\mathrm{avg}} - v_{\mathrm{avg}}^2/2)}$ and the adaptive $p_\star$ estimate of Toro's hybrid scheme, both derived for one ideal gas. They are given the arithmetic mean of the two face values. This is a wave-speed estimate, not a definition of the steady state: the converged solution is the root of a residual assembled from the conservative fluxes, and those carry each side's own equation of state. The HLLC branch needs no such average, its Davis–Einfeldt speeds taking one sound speed per side already.

5.5 Reduction to the atomic case

An atomic run must reproduce the constant-γ arithmetic to the bit, and the formula cannot be relied on to do it: with $n(H_2) = 0$, (9) evaluates $1 + 1/1.5$, which is not the same double as $5/3$. The reduction is therefore structural. A run with molecular chemistry off leaves every map on the legacy expression verbatim, and inside a molecular run a cell with exactly zero H_2 takes the same expression. Every non-molecular case of the regression matrix is byte-identical across the change.

6. The element budget the carriers spend, and what the stationary solve does with it

Everything above treats one cell's chemistry. This section is about the bookkeeping that holds across cells and across the two solvers, because that is where the stage-2 work of 2026-09 landed and because it changes what a molecular number in this memo means.

6.1 One map, three contexts

Every stoichiometric coefficient H_2 carries is now read in exactly one place, `nuclei_per_particle` of `src/modules/lower_atmosphere/element_inventory.f90`: two hydrogen nuclei for H_2 and H_2^+, three for H_3^+, one hydrogen and one helium for HeH^+. `element_inventory_of_cell` builds from it the full inventory of a cell grouped into what the carriers hold, what is locked, what the reservoir specifies and what the closure supplies, together with the budget each side spends; `element_inventory_of_candidate` does the same for a trial state of the Newton against frozen element densities; `EXHALE_INVENTORY_REPORT=1` prints either at the operator's entry.

The map exists because the coordinate box the carrier solve used to be limited by was the wrong feasible set. Five separate box sides, one per carrier, let two carriers of the same element each spend the whole budget: a cell sitting at both of its hydrogen-bearing ceilings breached the shared hydrogen budget by exactly 1.0 of the available hydrogen (MEASURED by the map on a coupled state). The marching path never showed it, because there `limit_to_element_budget` enforces the shared sums; what had no shared statement was the coupled stationary solve, whose residual compared a count of cells.

6.2 The constraint replaces the box

Since 2026-09-09 the trial step of the stationary solve is limited by the shared rows themselves,

$$2n(H_2) + n(H^+) + n(OH) + 2n(H_2O) \;\le\; n_H^{\mathrm{avail}},$$

and its oxygen and carbon counterparts, in place of the five corners (each corner is implied by the row, so nothing is lost and the direction along the constraint surface is gained). A nonlinear feasibility screen runs at every candidate, and an infeasible start is repaired by a restoration phase before the step is taken (on the coupled element-and-carrier fixture, 52 trials restored over 1165 cell-steps, worst violation $1.630 \to 2 \times 10^{-16}$; READ from the changelog). Two switches expose the change for measurement: `EXHALE_ELEMENT_CONSTRAINT_ROWS=0` restores the five corners and

EXHALE_ELEMENT_WRITE_BACK governs the element-conserving write-back below; no input key, output column or unknown layout changed.

The write-back is the part that mattered numerically. Each Newton iterate writes its carrier densities back into the species vector, and the old write-back did not return the element totals it had been handed: inside one coupled carrier solve the He/H nucleus ratio drifted from 1.39×10^{-11} to 2.016×10^{-1} of the value the run was given over 79 iterates, while every carrier box reported a breach of exactly zero, so no gate the code had could see it. The element-conserving write-back holds the same ratio at 1.39×10^{-11} at all 68 iterates of the same fixture (MEASURED, N4b). This is bookkeeping, not chemistry, but a composition whose helium content has moved by 20% is not the composition any rate in this memo was evaluated for.

A clamp that creates nuclei, still in the tree. When the proportional share of a hydrogen write-back leaves a cell whose local stages hold nothing (held ≤ 0), the remainder is deposited as

$$f(\mathrm{H\,I}) = \max(\mathrm{rest}, 0)/n_{\mathrm{d}},$$

and with rest < 0 the carriers keep what they hold while the closure cannot go negative, so the written element total exceeds the handed one by |rest|: nuclei are created. The same shape appears in the oxygen and carbon deposits of the same routine (diffusive_photochemistry.f90). Only element_census_verify under EXHALE_ELEMENT_ASSERT = 1 or 2 sees it, and that is off by default. The finding is recorded and the clamp is unchanged; it is not a defect of a rate but of the reconciliation around it.

6.3 H_2 is transported; the molecular ions are eliminated

The carrier option transports H_2 alone and closes H_2^+, H_3^+ and HeH^+ locally, and that split is now measured rather than asserted. On the resolved gradient length of the hot-Uranus state the product of the advection time and the chemical decay rate, $t_{\mathrm{adv}}\lambda$, is 3.68 at the base, 4.0×10^{-3} at the front and 0.82 in the wind for H_2, while H_2^+, H_3^+ and HeH^+ decay three to twelve decades faster than transport everywhere (READ from the changelog). H_2 is therefore the only molecular species whose local balance the flow can break on this grid, and none of the three ions is proposed for promotion. The parameterization removes the conservation modes by itself: the neutral stage closes each element.

The two-root H_2 front does not reproduce once H_2 is transported. With the eliminated closure, seeding the H_2 partition from ten different starting points found two distinct roots at the front. With H_2 among the transported carriers, $x(H_2)$ at the front is bitwise identical over the same ten seeds, because the transported value replaces the seed in the H_2 row. The H I partition still shows one root on the carrier fixture and two on the element-and-carrier fixture (9 seeds and 1, 1.05×10^{-1} apart); whether the second is a distinct branch or a slow iteration is not settled. Both readings are from the branch-discovery option EXHALE_RESID_BRANCH_REPORT=1 of the residual contract (N5).

6.4 The upstream caloric energy in the _adv residual

The advection-corrected post-process does not carry the molecular columns, but its energy equation is no longer molecule-free. The upwind term of the _adv energy residual receives the upstream cell's specific internal energy

$e_{\rm up} = E(x_{\rm H_2,up}, T_{\rm up})/\mu_{\rm up}$ formed by the caller, and the caloric branch is taken when *either* side of the interface carries H_2, so an atomic cell sitting above a molecular one no longer loses the incoming rovibrational energy; both sides atomic gives the monatomic expression verbatim (`post_process_adv.f90`, `T_equation.f90`, `ion_cell_state.f90` fields `x_h2_up` and `e_up`). A larger defect was fixed in the same path: `x_h2_cell` had been assigned only inside the stationarity loop and read by the temperature loop for every cell, so the whole molecular layer was solved at the H_2 fraction of the outermost ghost (1.5×10^{-7} on `mol_base_handoff`, 2.3×10^{-5} on `mol_carrier`, against 0.84 at the base).

The energy convention is unchanged and nothing is counted twice: the caloric equation of state carries the bound rovibrational ladder only, and the 4.478 eV bond lives in the formation reservoir of `molecular_reaction_heat.f90`. The consequence for anyone reading an `_adv` file of a molecular run is large. Between the two defects, `T_adv` moved in 469 and 467 of 504 rows on `mol_base_handoff` and `mol_carrier`, by up to 94.2% (218 → 424 K at $r = 1.094$) and 103.2% (169 → 343 K), in both cases a warming of the molecular layer, which is what carrying $\frac{5}{2}kT$ per H_2 instead of $\frac{3}{2}kT$ does (numbers READ from the changelog). The molecular `_adv` goldens were refreshed at the 2026-09-10 gate. The product is still not a molecular stationary solution, and the writer says so in its own header.

7. Validity ranges

Historical model status (2026-10-27). The comparisons and table below retain their original model states. In particular, the H_2 photoabsorption row describes an older Yan-only fit; current `sigma_H2` uses measured tables and a high-energy Yan tail. The statements that helium vibrational collision rates are unavailable and the quoted critical-density comparisons are also stale for the current collider coefficients. These entries require a new evaluation with the current cross sections, collider mixture, and relaxation rates before they are used as current validity judgments.

Table 3 collects every range in one place: what each treatment stands on, and what EXHALE asks of it.

One of those rows is about the closure and not about a coefficient, and it is the one that fails. Everywhere in this document H_2 is a root of a local algebraic balance, recomputed from scratch in each cell by `ioniz_eq`. That is a statement about time scales, not a property of the network: it holds where the chemistry is fast against the flow and nowhere else. Evaluated on our own converged solution it does not hold across the H_2 front: the term the closure drops is two decades larger than the whole net chemical rate there, so the front position and the H_3^+ extent that follow from it are properties of the closure as much as of the chemistry. The transported alternative exists; it is off by default while the question of whether the transported problem has a steady state on that planet is still open.

The transported operator has a validity range of its own, and it is numerical rather than chemical. Its advection was first-order donor-cell until 2026-09-03, carrying a numerical diffusivity $|v|\Delta r/2$ that on the 500-cell grid of the hot Uranus is 8 to 22 times the molecular diffusivity of H_2 over 1.15–1.50R_p, so the front it produced was set by the scheme as much as by the flow. It is now second order (van Leer limited slope added as a deferred correction), which on the coarse grid reproduces what doubling the grid gave. The residual front motion after that fix is grid-converged, so it is the state relaxing and not the discretization; but the state is not steady, the steady solver stalls on the energy row inside the front, and no number from a transported run is a converged one yet.

And the way the transported problem was being closed against the wind was itself wrong, in sign. Until 2026-

09-03 the steady route alternated a wind solve at a frozen H_2 partition with a carrier relaxation at a frozen wind. That alternation does not converge and cannot: its acceptance gate stood below the smallest move the transport operator can make, the fixed-wind steady state it relaxed to sat on the hydrogen element ceiling over a quarter of the grid, and its convergence measure (the carrier row divided by $n_{\mathrm{H}}(|v|+c_s)/\Delta r$) understated the row's own imbalance by a factor $10^{2.7}$ to $10^{6.4}$, so a state carrying 0.60 of its own terms at the front was reported at 1.6×10^{-4}. Worse, the splitting moved the H_2 front OUTWARD at 83% of the local gas speed, whereas solving the carriers and the wind together moves it INWARD (from $x_{\mathrm{H_2}} = 0.5$ at 1.142 to $1.062\,R_p$) because H_2 costs particles, fewer particles at the wind's own pressure means a hotter cell, and a hotter cell destroys H_2. A fixed wind cannot see that feedback. The carrier row now carries the same convergence statement as every other row (its own largest terms), and `Coupled carrier solve` puts $n(H_2)$ among the Newton unknowns so that the two halves are solved at once. The coupled solve improves the residual by a factor three and the front's own carrier imbalance by four, and then stagnates with the molecular layer 0.6 to 4% out of carrier balance, so the closure question above stands: the transported problem still has no converged steady state on that planet, and the option stays off by default.

Everything else in this section is about how well a rate is known; this row is about whether the equation is the right one. The evaluated column is the $x_{\mathrm{H_2}} > 0.01$ layer of §1.2 unless stated, with the caveats recorded there about relaxation snapshots.

Table 3: What each treatment stands on and what it is asked to do.

Treatment	Range of the data	Range evaluated	Verdict
H_2 LTE line emission	Vibrational thermalization needs $n \gg n_{\mathrm{crit}}(v{=}1)$; A_{10}/γ_{10} of Hollenbach & McKee (1979) is 2–6 $\times 10^6$ cm^{-3} (H_2 collider) or 5–8 $\times 10^4$ cm^{-3} (H collider) at 900–1350 K	$n(H_2) = 4.5 \times 10^9$–5.2×10^{12} cm^{-3}	Inside everywhere, by three to eight decades. No helium collision rate is published in either source, so a helium-dominated gas is an extrapolation: one that understates the de-excitation rate
H_2O and CO band cross sections	Tabulated 50–2000 K, clamped outside	874–1221 K in the molecular layer	Inside. Both molecules are largely dissociated above 2000 K in any case
R15 three-body recombination, $k_1(H_2) = 2.8 \times 10^{-31} T^{-0.6}$	Cohen & Westberg (1983), recommended 50–5000 K; ± 0.2 in $\log k$ at 300 K rising to ± 0.4 at 5000 K	874–1221 K	Inside

Table 3 (continued)

Treatment	Range of the data	Range evaluated	Verdict
R12 thermal dissociation, $k_{\rm rec}/K_{\rm eq}$	Inherits R15's range; $K_{\rm eq}$ tabulated 100–20000 K and clamped	874–1221 K	Inside. Against the Baulch et al. (1992) fit the ratio is 0.97–1.18 over that fit's own 2500–8000 K and departs only below it
Third body M = everything	Measured for M = H_2, H, Ar, Xe (Breshears & Bird 1973); no helium coefficient exists	Total heavy-particle density	H_2-dominated base: inside. Above the front, M is atomic H, a factor 2–5. Helium: unquantified
Lyman–Werner $\sigma_{\rm diss}$, column axis	Tabulated 10^{12}–5×10^{21} cm^{-2}, overlap included; cross-checked against the Meudon PDR code to 10^{21}	up to 4.36×10^{21}, and 5.31×10^{21} in transient	Inside the axis. The transient excursion is 6% past its top and is clamped there; the second-code cross-check stops at 10^{21}
Lyman–Werner $\sigma_{\rm diss}$, Doppler axis	Not an axis: the profiles are Voigt with thermal $b=(2kT/m_{\rm H_2})^{1/2}$ at the tabulated T	thermal b = 2.79–4.48 km s^{-1}	Inside. `h2_doppler_parameter` evaluates the same expression, so table and code agree by construction
Lyman–Werner $\sigma_{\rm diss}$, level distribution	LTE at the tabulated T = 700–3200 K; the CLOUDY runs on the same grid return departure coefficients of 1.000	874–1221 K, $n_{\rm H}$ up to 1.1×10^{13} cm^{-3}, collisional	Inside. Turbulent broadening and any non-thermal population are not carried
Lyman–Werner trapping ratio $p_{\rm eff}/p_{\rm single}$	CLOUDY escape probabilities in a plane-parallel slab, illuminated on one face and closed on the other	a spherical layer, thick inward and open outward	**Outside on the geometry.** The thick-inward side is the one that matters, but the open outward side is not what was computed; the size of the mismatch is unquantified

Table 3 (continued)

Treatment	Range of the data	Range evaluated	Verdict
Band fraction A from DB96 eq. (39)	Their own ceiling, §4.3 eq. (29), is $W_{\max} \approx \ln(1110/912) \approx 0.2$, i.e. $A = 1$	asymptote 1.0843 at $b = 2.79$, 1.1050 at 4.48 km s^{-1}	**Above the authors' own cap** by 8.4–10.5%; clamped at 1 in the code
H_3^+ LTE emission $E(T)$	MiZATeP cooling function (Mizus et al. 2017), 1 K nodes 20–5000 K; held above	874–1221 K	Inside (the layer temperatures were taken with the Miller et al. 2013 fit, which this function replaced on 2026-10-10; the range lies well inside the table either way)
H_3^+ departure factor $s(T, n_{H_2})$	Table 6 grid T = 300–5000 K, $\log_{10} n_{H_2}$ = 6–14; edge held above 10^{14} (s = 0.9985–1), $s \propto n_{H_2}$ below 10^6	T = 874–1221 K, $n_{H_2} = 4.5 \times 10^9$–$5.2 \times 10^{12}$	Inside; s = 0.44–0.996 across the layer, both values stale (§2.4)
`q_h2_equilibrium`	Koskinen et al. (2022) Eq. (11), validated against NASA CEA at 1000–2500 K; thermochemical equilibrium only	Seed and base-composition use only	Inside as a seed. Not applicable to an irradiated wind: it has no photodissociation

Table 3 (continued)

Treatment	Range of the data	Range evaluated	Verdict
Photoelectron energy degradation in $H/H_2/He$ (Dalgarno et al. 1999, `electron_energy_degradation.f90`)	Fits stated for ionization fraction $x \leq 0.1$; tables at $E_0 = 30$–1000 eV, clamped outside; helium fixed at $n(\mathrm{He}) = 0.1[n(\mathrm{H}) + 2n(\mathrm{H_2})]$ and no metal at all; H_2 in its ground vibrational level, so $T \lesssim 2000$ K	$x = 10^{-16}$–5×10^{-5} in the layer and $\to 1$ in the wind above it; $E_0 > 30$ eV (`E_sec_ion`); 874–1221 K	Inside on x in the layer, on E_0 and on T. **Outside on x in the ionized wind**, where eq. (14) is extrapolated above 0.1 with a neutral-fraction closure that forces the $x = 1$ endpoint; that extrapolation is accepted on a measurement against Shull & van Steenberg's Table 1 rather than on the closure (§118.6 of `Update_EXHALE`). **Outside on helium** at He/H $\sim$ 1–10, by one to two orders in the direction of more helium: the branching between targets is renormalized onto the cell's own densities, but the total ionization budget and the heating fraction still carry the sources' own 0.1. **Outside on metals**, whose electrons enter the ionized fraction although neither source has a metal in its gas

Table 3 (continued)

Treatment	Range of the data	Range evaluated	Verdict
H_2 vibrational energy: where it goes (Hollenbach & McKee 1979; Burton et al. 1990, `h2_vibrational_relaxation.f90`)	Rate coefficients of HM79 eq. (6.29) fitted "good to 20% for $T \gtrsim 500$ K"; colliders H and H_2 only (Burton et al. add electrons); no helium rate in either source; $E_* = 2.0$ eV is an adopted value for *photon* pumping	$T = 874$–1221 K; colliders are H_2, H and He with $n(H_2) = 4.5 \times 10^9$–5.2×10^{12} cm^{-3} against $n_{cr}(v{=}1) = 5 \times 10^4$–$1 \times 10^6$ cm^{-3}	Inside on T and, by three to eight decades, on density: the quench fraction is 1.0000 throughout the layer. **Outside on helium**, which is left out of the collider sum for want of a rate: an omission that lowers the quench fraction, so it errs against claiming the energy as heat. Applying E_* to electron-impact excitation of B and C is an assumption: same upper states and cascade, different distribution over upper vibrational levels
`sigma_H2` photoionization	Yan et al. (1998) Eqs. 17–19, $E \geq 15.4$ eV, ground vibrational state, resonances smoothed	the full stellar spectrum above threshold	Inside in energy; the vibrational-state assumption is unquantified
`frac_H2_dissociative_ionization`	Chung et al. (1993) Table II, 18.076–124 eV, room-temperature target, stated uncertainty ±4–5%	the full stellar spectrum above 18.076 eV	Inside up to 124 eV; **held flat above it**, where the source has no data. Above ~ 47 eV the measured $\sigma(H^+)$ counts the protons of the double-ionization channel as well ($\sim 20\%$ of it by 80 eV), so the proton source is right and the co-product assignment is not
$He(2^3S) + H_2$ ionization	Cohen & Lane (1977) via Garcia Munoz (2025), tabulated 500–10^4 K	layer temperatures, evaluated unclamped outside	Inside over the molecular layer

Table 3 (continued)

Treatment	Range of the data	Range evaluated	Verdict
H_2 solved as a LOCAL chemical equilibrium (the closure itself)	Valid where the chemistry is fast against the flow, i.e. where the Damköhler number $\tau_{\rm adv}/\tau_{\rm chem} \gg 1$	The whole domain, in every run that does not set `Molecular carrier transport: True`	**OUTSIDE across the H_2 front.** Measured on the converged He/H = 0.0793 hot Uranus, the omitted advection term $\lvert r^{-2}\partial_r(r^2 v n_{\rm H_2})\rvert$ is 44 and 117 times the whole net chemical rate at 1.05 and $1.15\,R_p$, and only at $1.25\,R_p$ does the chemistry win (0.6). Inside beyond about $1.25\,R_p$ and at a base where $v = 0$. The transport operator that removes the restriction is `diffusive_photochemistry.f90`; it is default off for a molecular-only run

8. Open items

- **No (v, J) resolution, and why it has not been added.** Carrying the ladder as transported species would mean 302 further continuity equations, the 1833-line radiative network as a level-coupling matrix instead of a summed emissivity, and a state-to-state collisional set that the code does not have and that none of its current sources supply. Against that cost, the place it would most obviously buy something (the self-shielding) is now supplied from outside by a line-by-line calculation that carries the ladder for us (§2.3), so the transported ladder would buy consistency rather than a missing number. The trade is recorded here rather than settled.

- **Self-shielding at our temperature: DONE, and it did not transfer.** This item asked for $f_{\rm shield}$ on a 10^3 K collisionally equilibrated population at the columns of Table 3. It was computed (§2.3), both published fits failed against it, and the rate now reads the calculation.

- **Line overlap: DONE; the geometry of the trapping is what is left.** The successor item (that above $N_{\rm overlap}(T)$ the tabulated calculation was itself an upper bound, because it shielded one line at a time) is closed. The table is now built with every Lyman and Werner line absorbing every other line's beam, `h2_shielding_overlap_column` is gone, and the only column boundary left is the top of the grid (§2.3).

What replaces it is narrower: the trapping ratio $p_{\rm eff}/p_{\rm single}$ is still CLOUDY's, computed in a plane-parallel slab illuminated on one face and closed on the other, while our layer is spherical and open outward. The thick-inward side is the side that dominates, but the size of the mismatch has not been quantified.

- **An internal inconsistency in the published DB96 fit.** DB96 cap the overlap-corrected equivalent width of the pumping lines at $W_{\rm max} \approx \ln(1110/912) \approx 0.2$ (their §4.3, eqs. 29–30), i.e. at a band fraction of exactly 1. Their closed-form integral, eq. (39), has the asymptote $1.05(1+0.0117\,b_5)$ as $N_{\rm H_2} \to \infty$, which at our Doppler parameters is 1.0843 ($b = 2.79$ km s^{-1}) to 1.1050 (4.48 km s^{-1}): 8.4–10.5% above their own ceiling. Their remark below eq. (39) is only that it "corresponds to an equivalent width $W \approx \Delta \ln \nu \approx 0.2$ in the limit $N_2 \to \infty$". The overshoot is a property of the published fit, not of our use of it, and is the same whether the band fraction is taken from eq. (39) or from integrating $f_{\rm shield}$ on the grid. `fuv_lw_photon_field` clamps it at 1 so the transmission handed to the continuum absorbers of the same interval cannot go negative.

- **The `mol_ir_bands` regression case does not test what its name says.** It sets `Base IR field: True` and `Molecular IR bands: True` but leaves `Oxygen chemistry` off, as its `EXHALE_resolved.out` confirms, so there are no H_2O or CO carriers. Measured on `output/Cooling_breakdown.txt`, columns 13 and 14 are exactly zero in all 504 rows, while column 12, the H_2 line channel, is nonzero in all 504 and peaks at 2.148×10^{-8} erg cm^{-3} s^{-1} at $r = 1.0059\,R_{\rm p}$, $T = 1248$ K. The zeros are structural rather than numerical: with $n(H_2O) = n(CO) = 0$ the two net-rate functions return 0 on their own guard. The case gates the H_2 line channel and nothing else, and a case that exercises the H_2O and CO paths would have to add `Oxygen chemistry: True`. *Note also* that the golden directory for this case carries `Hydro_ioniz.txt`, `Ion_species.txt` and their two `_adv` twins, plus `run.log`; `Cooling_breakdown.txt` is not snapshotted, so the infrared columns are not under regression control at all.

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